Project 4 Team 2 — Analytical Report

Ride-Share Demand Analysis

Executive Summary

This analysis examined 228,851 Chicago taxi trips from 2023 to understand demand patterns, fare dynamics, and profitability drivers. Key findings include strong weekday demand peaking in late afternoon/early evening and early morning, significantly higher tipping rates for credit card payments, and the most profitable time for drivers being Tuesday at 2 AM. We recommend optimizing driver availability during identified peak times and promoting credit card usage to maximize earnings.

1. Introduction and Objectives

This report aims to analyze Chicago taxi trip data for the year 2023 to identify patterns in ride-share demand, understand fare and tipping behaviors, and assess company performance. The primary objective is to answer business-critical questions such as peak demand times, factors influencing profitability, and operational insights that can inform strategies for taxi companies and individual drivers. Understanding these dynamics is crucial for optimizing driver deployment, implementing effective pricing strategies, and ultimately increasing efficiency and profitability within the competitive ride-share market.

2. Data and Methodology

The analysis utilized the `chicago_taxi_trips` dataset from `bigquery-public-data.chicago_taxi_trips.taxi_trips` covering the entire year of 2023. Due to BigQuery memory constraints and to ensure statistical validity for payment-type analysis, a stratified random sample was initially pulled, limiting each `payment_type` to 50,000 records. This resulted in a dataset of 228,851 valid trips.

- **Data Cleaning Decisions:**
- Trips with `fare`, `trip_seconds`, or `trip_miles` less than or equal to zero were excluded to remove invalid or incomplete trip records.

- The `trip_start_timestamp` column was converted to a datetime object, and additional time-based features (`hour`, `day_of_week`, `month`) were extracted for temporal analysis.
- Missing values were observed in `pickup_community_area` (5,483 records), which were noted as a limitation for precise geographic analysis.

3. Key Findings

Finding 1: Overall Metrics Highlight Significant Market Activity

The analysis of 228,851 valid taxi trips in 2023 reveals a total fare revenue of **$5,181,896.88**, with an average fare of **$22.64**. The typical trip lasted **22.26 minutes** and covered an average distance of **6.98 miles** (Section 2).

Finding 2: Dual Peak Demand Periods Throughout the Day

Trip volume analysis by hour of day shows two distinct peak periods: a primary peak in the **late afternoon/early evening (4 PM to 6 PM)**, consistent with end-of-workday commutes, and a secondary peak in the **early morning (1 AM to 2 AM)**, likely reflecting late-night activities and airport transfers (Section 3).

Finding 3: Weekdays Drive Higher Demand, with Friday as the Weekly Peak

Demand varies significantly by day of the week, with **weekdays showing substantially higher trip volumes compared to weekends**. **Friday stands out as the busiest day**, followed by other weekdays, suggesting strong commuter and end-of-week social activity (Section 4).

Finding 4: Seasonal Fluctuations in Trip Volume

Monthly trends indicate seasonal patterns, with **higher trip volumes observed during the summer months (June, July, August)** and a noticeable dip towards the end of the year, potentially influenced by holiday periods or weather changes (Section 5).

Finding 5: Fare Distribution Shows a Predominance of Lower Fares with High-Value Outliers

The majority of taxi fares are concentrated **below $50**, indicating frequent short-to-medium distance trips. However, the distribution also reveals a long tail with significant outliers, such as

the **absolute maximum fare of $490.00,** suggesting occasional very long-distance or premium trips (Section 6).

Finding 6: Major Companies Dominate Volume, While Smaller Ones May Command Higher Average Fares

Taxi Affiliation Services, Flash Cab, and City Service account for the highest number of trips, indicating their dominant market share. Interestingly, while smaller companies have fewer trips, some like 5 Star Taxi show a tendency towards **higher average fares**, potentially serving niche markets or premium services (Section 7).

Finding 7: Credit Card Payments Yield Significantly Higher Tip Percentages

Analysis of payment types shows that **Credit Card payments result in a considerably higher average tip percentage** compared to Cash or Mobile payments. This suggests that customers paying with cards are more inclined to tip and tend to tip a larger proportion of their fare (Section 8).

Finding 8: Strong Linear Relationship Between Fare and Trip Distance

There is a clear **positive linear relationship between fare and trip distance**, meaning longer trips generally incur higher fares. However, the scatter in the plot indicates that other factors, such as duration, time of day, or specific routes, also influence the final fare amount (Section 9).

4. Crown Metric Analysis

Question: What is the most profitable hour-and-day combination for a Chicago taxi driver?***Answer:*** The absolute best time for a Chicago taxi driver to maximize average trip value (fare + tip) is **Tuesday at 2:00 AM.**

Supporting Evidence:

- **The Math Angle:** Data indicates that a trip initiated on Tuesday at 2:00 AM yields an average total value of **$39.55** (composed of an average fare of $33.35 and an average tip of $2.16). This figure represents the highest average value across all hour-and-day combinations observed in the dataset.
- **The Real-World Angle:** This peak profitability at an unconventional time can be attributed to several real-world factors. At 2:00 AM on a Tuesday, public transportation options like 'The L' often run less frequently. Passengers needing taxis at this hour are

frequently taking very long trips, such as those arriving on delayed late-night flights at O'Hare airport who require travel to distant suburbs. These long-distance, time-sensitive journeys result in substantially higher fares and often more generous tips (Section 10).

5. Limitations

Data Sample Size: The initial analysis was based on a stratified sample of 228,851 trips, not the entire 2023 dataset (which would be over 5 million records). While designed for balance, this may not capture all nuanced patterns present in the full data.

Missing Geographic Data: A notable number of trips (5,483) are missing `pickup_community_area` information, which limits the ability to perform granular geographic analysis or understand demand patterns at a community level.

S*ingle-Year Focus*: The analysis is restricted to 2023 data, preventing year-over-year trend analysis or identification of long-term shifts in demand.

Exclusion of External Factors: The current dataset does not include external factors such as weather conditions, major city events (concerts, sports), public transit disruptions, or economic indicators, all of which could significantly influence demand and pricing.

No Driver-Specific Data: The analysis focuses on trip characteristics rather than individual driver performance or behavior, which could offer insights into optimizing personal earnings.

6. Recommendations

Optimize Driver Scheduling: Taxi companies and drivers should consider prioritizing availability during identified peak hours, particularly **Friday afternoons/evenings and Tuesday at 2 AM**, to capitalize on higher demand and profitability.

***Promote Credit Card Payments:* *** Actively encourage the use of credit card payment options (e.g., through incentives or prominent display of card readers) to potentially increase average tip revenue for drivers.

Target Specific Routes/Services: Investigate the nature of high-value early morning Tuesday trips (e.g., airport service) and consider offering specialized, premium services or dedicated drivers for these routes.

Address Data Gaps: Future analyses should aim to address the `pickup_community_area` data gap to enable more precise location-based demand forecasting and service optimization.

***Dynamic Pricing Strategy Review*:** Explore implementing or refining dynamic pricing models that adapt to real-time demand fluctuations, particularly during peak hours, to maximize revenue.

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7. Appendix — SQL Query Reference

```sql
SELECT * FROM (
   SELECT
      trip_start_timestamp, fare, tips, trip_total,
      trip_seconds, trip_miles, pickup_community_area,
      company, payment_type,
      ROW_NUMBER() OVER(PARTITION BY payment_type ORDER BY RAND ()) as row_num
   FROM `bigquery-public-data.chicago_taxi_trips.taxi_trips`
   WHERE EXTRACT (YEAR FROM trip_start_timestamp) = 2023
)
WHERE row_num <= 50000
```

This SQL query retrieves a stratified random sample of taxi trips from the `chicago_taxi_trips` BigQuery table for the year 2023. It partitions the data by `payment_type` and selects up to 50,000 random records for each payment type, ensuring a balanced representation for analysis, especially concerning tipping behavior. The selected columns include trip details like timestamp, fare, tips, duration, distance, pickup area, company, and payment method.